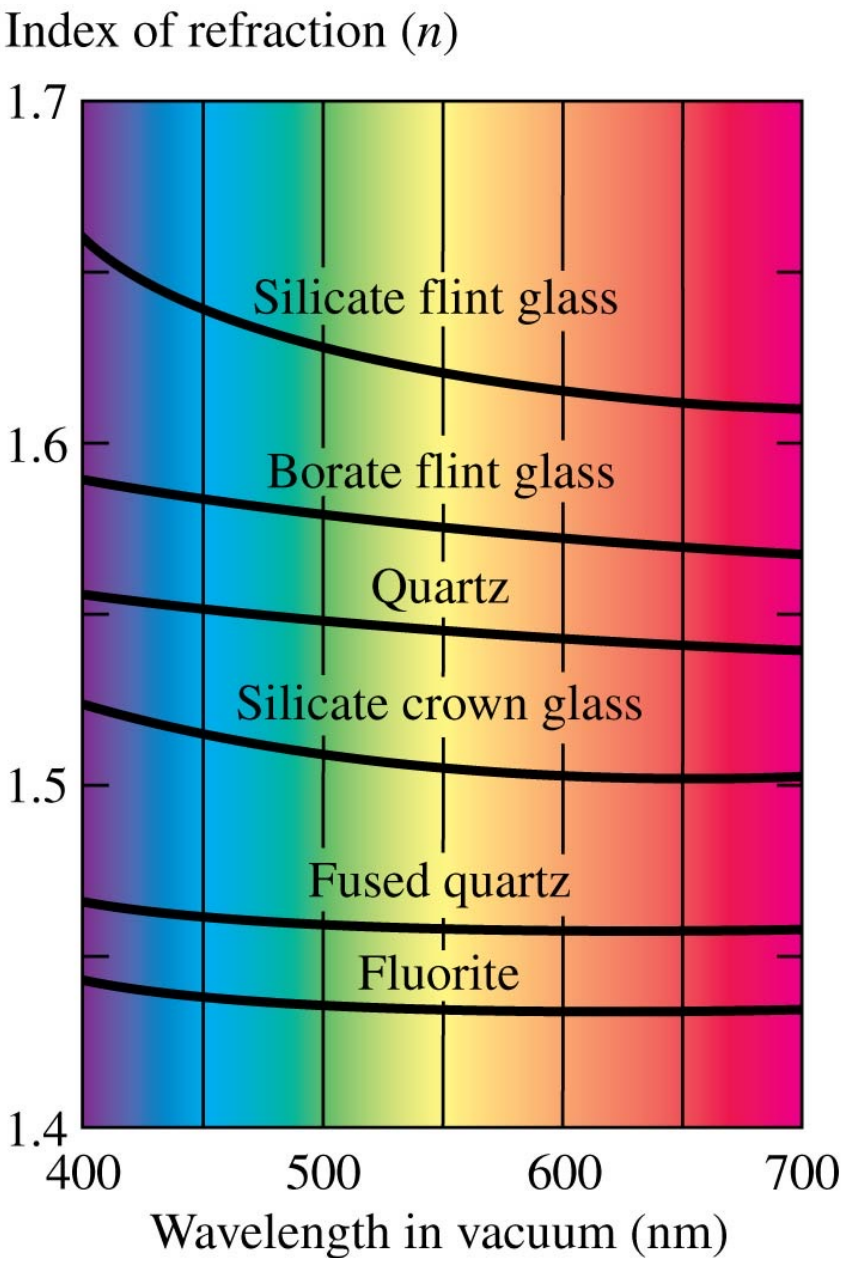
Dispersion

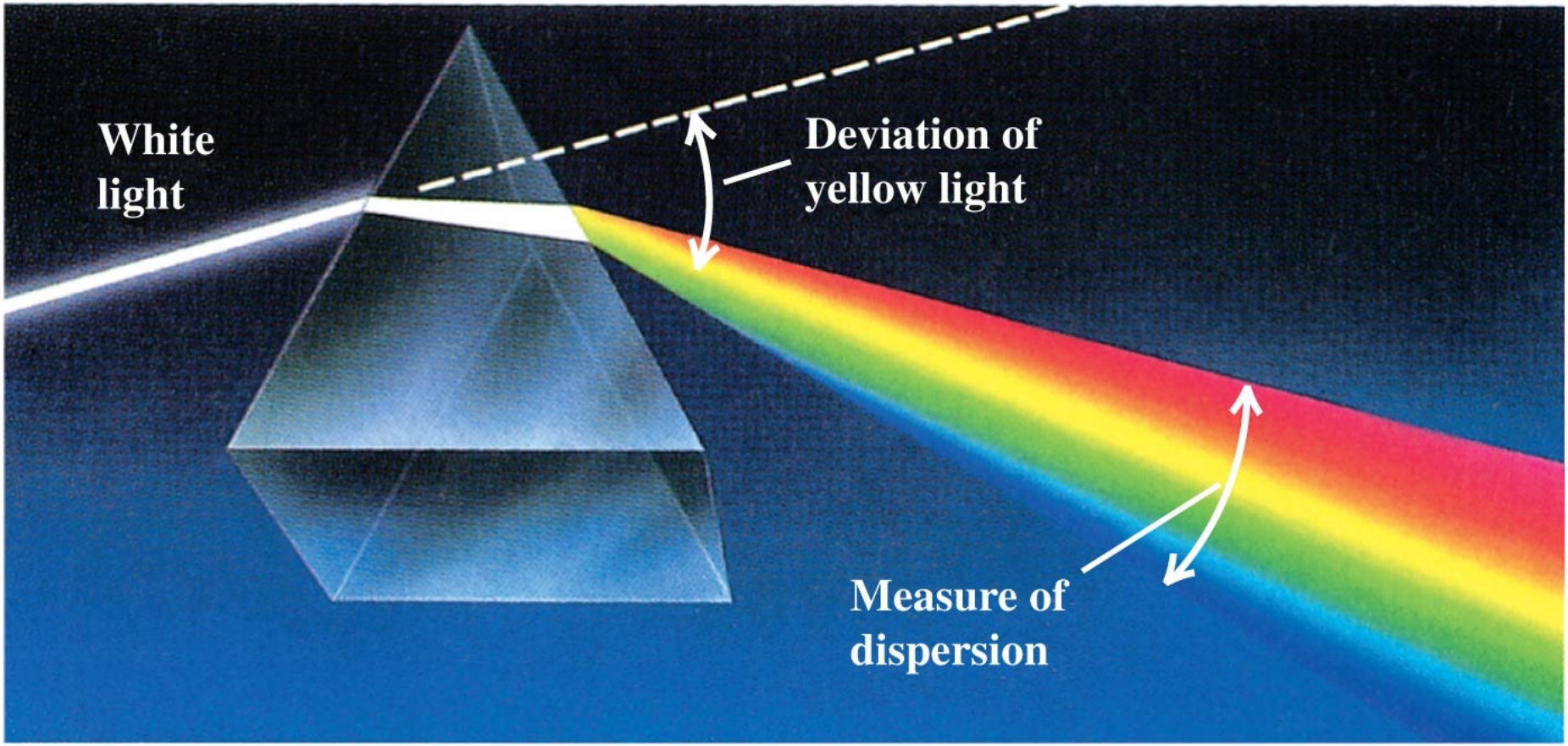
- The speed of light in vacuum is the same for all wavelengths, but the speed in a material substance is different for different wavelengths.
- The dependence of wave speed and index of refraction on wavelength is called **dispersion**.
- In most materials the value of *n decreases* with increasing wavelength and decreasing frequency.



© 2016 Pearson Education, Ltd.

Dispersion

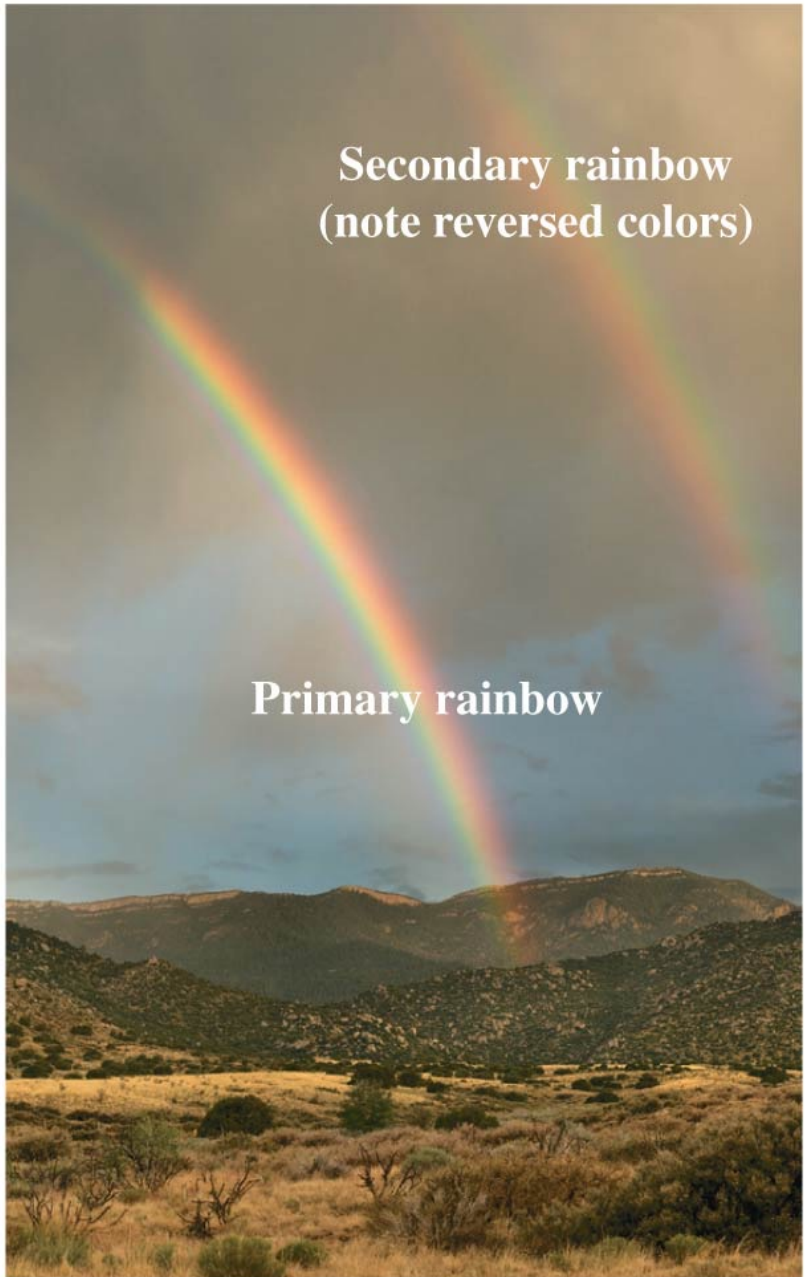
- Ordinary white light is a superposition of waves with all visible wavelengths.
- The band of dispersed colors is called a **spectrum**.



© 2016 Pearson Education, Ltd.

Question 23: Double rainbows

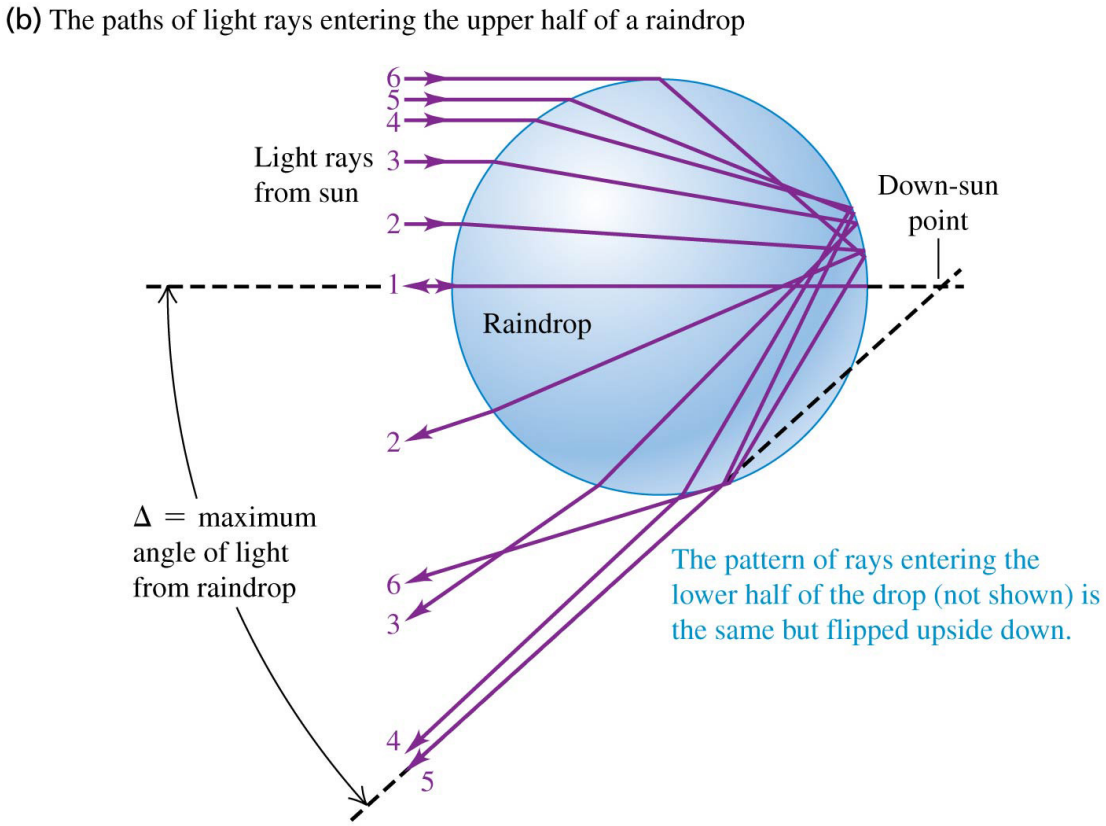
- Why the color spectra of primary and secondary rainbows are reversed with each other?
- Why the secondary rainbow is fainter than the primary?



© 2016 Pearson Education, Ltd.

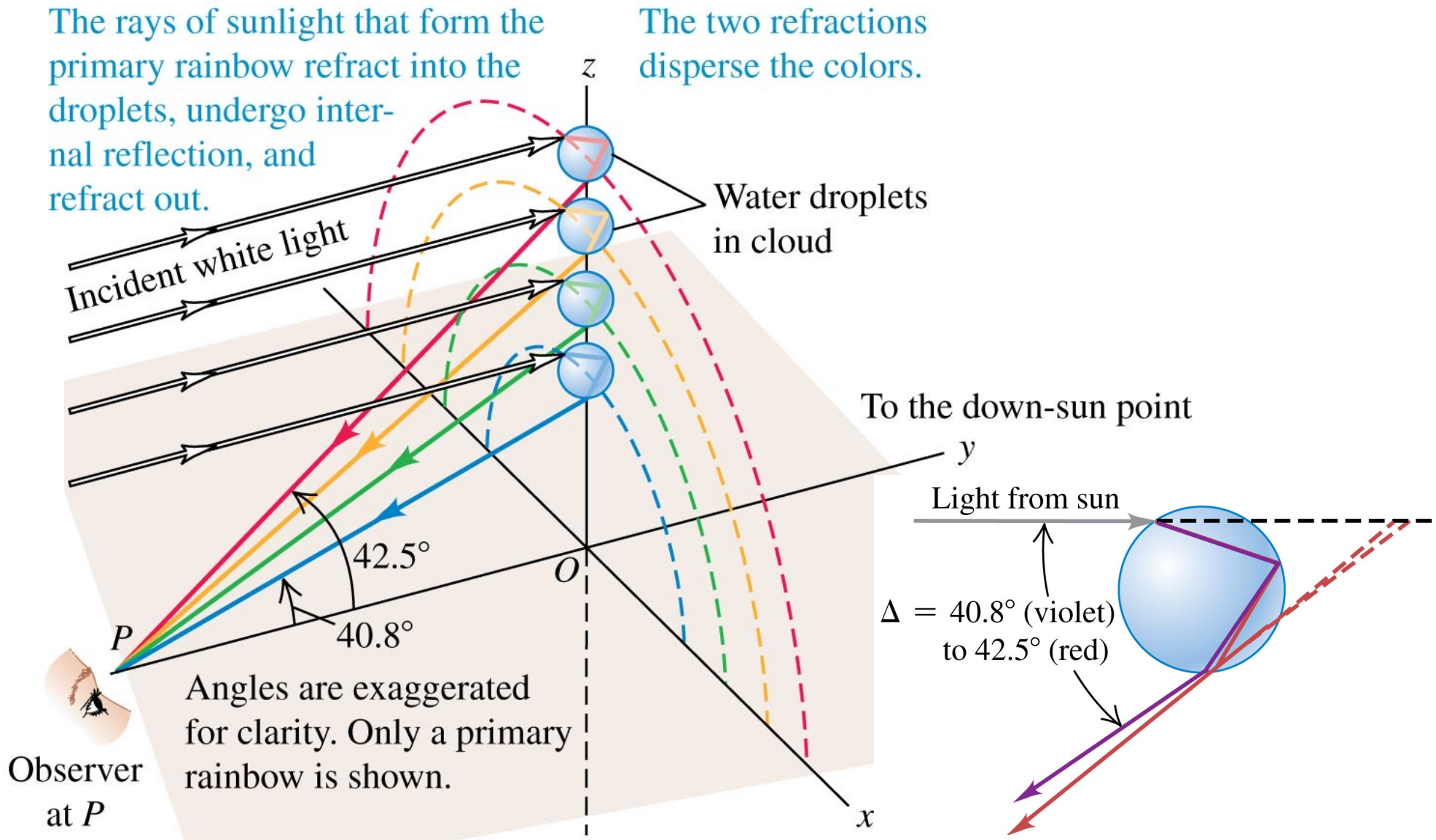
How rainbows form: Slide 1 of 3

- When sunlight enters a spherical water droplet suspended in the air, it is (partially) reflected from the back surface of the droplet, and is refracted again upon exiting the droplet.
- A light ray that enters the middle of the raindrop is reflected straight back.



© 2016 Pearson Education, Ltd.

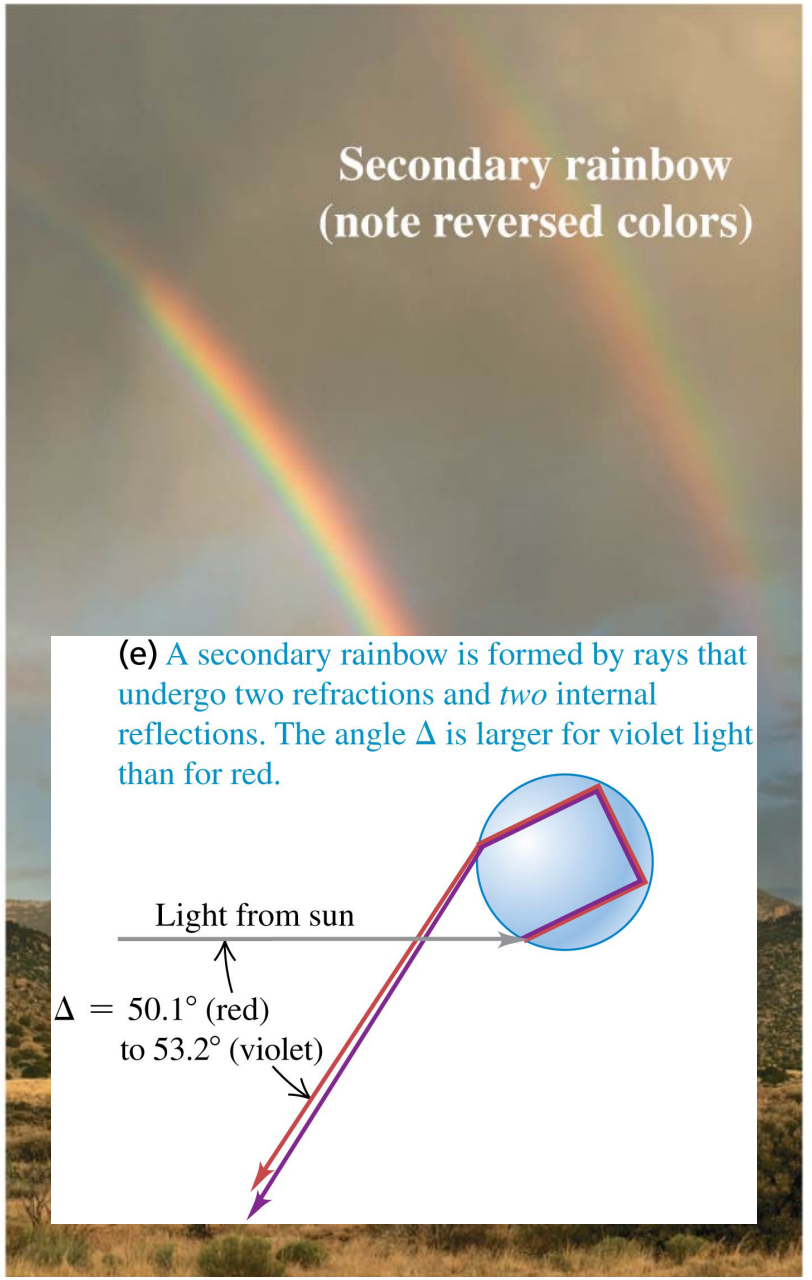
How rainbows form: Slide 2 of 3



© 2016 Pearson Education, Ltd.

How rainbows form: Slide 3 of 3

- In many cases you can see a second, larger rainbow.
- It is the result of two reflections from the back surface of the droplet.
- Just as a mirror held up to a book reverses the printed letters, so the second reflection reverses the sequence of colors in the secondary rainbow.



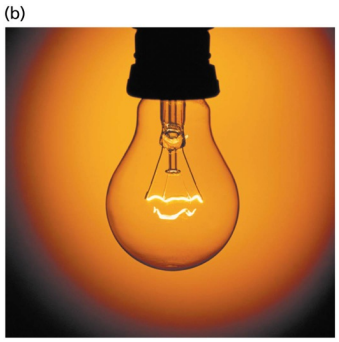
© 2016 Pearson Education, Ltd.

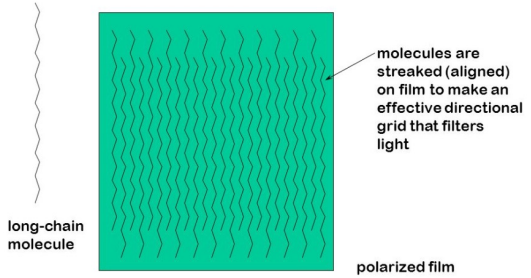
Polarization

- An electromagnetic wave is *linearly polarized* if the electric field has only one component.
- Light from most sources such as light bulbs is a random mixture of waves linearly polarized in all possible transverse directions; such light is called **unpolarized light** or **natural light**.
- A Polaroid **polarizing filter** can convert unpolarized light to linearly polarized light.

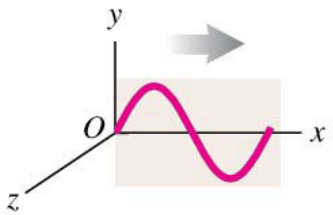
$$\vec{E}(x,t) = \hat{j}E_{\max}\cos(kx - \omega t)$$

$$\vec{B}(x,t) = \hat{k}B_{\max}\cos(kx - \omega t)$$

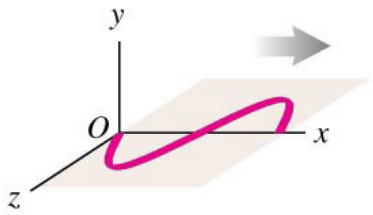




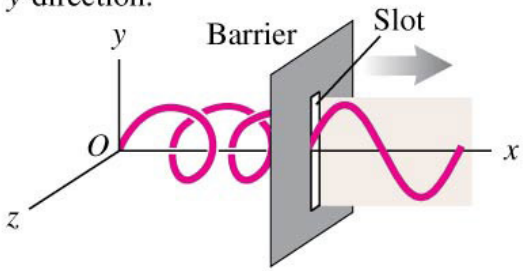
(a) Transverse wave linearly polarized in the y-direction



(b) Transverse wave linearly polarized in the z-direction

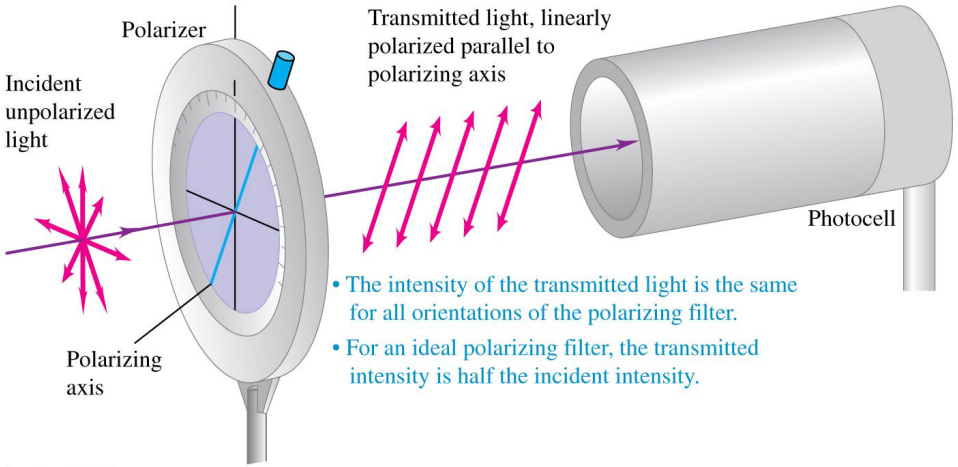


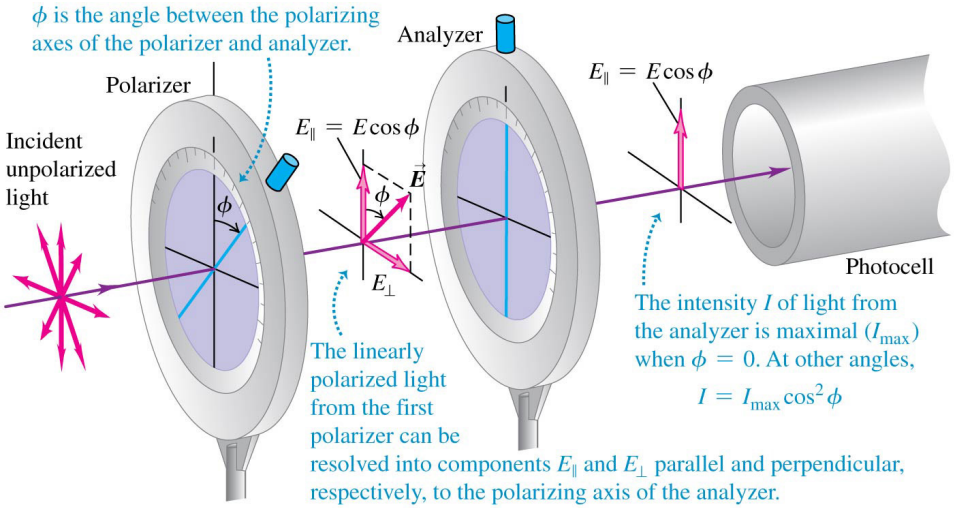
(c) The slot functions as a polarizing filter, passing only components polarized in the y-direction.



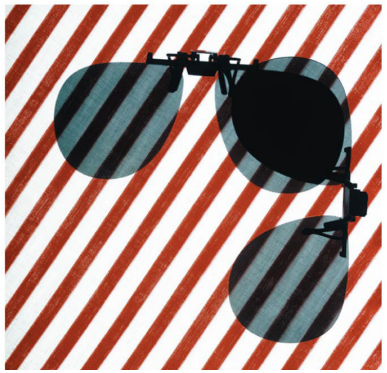
Malus's law

- When polarized light of intensity I_{max} is incident on a polarizing filter used as an analyzer, the intensity I of the light transmitted through the analyzer depends on the angle ϕ between the polarization direction of the incident light and the polarizing axis of the analyzer.









Intensity of polarized light passed through an analyzer

$$I = I_{max}\cos^2\phi$$

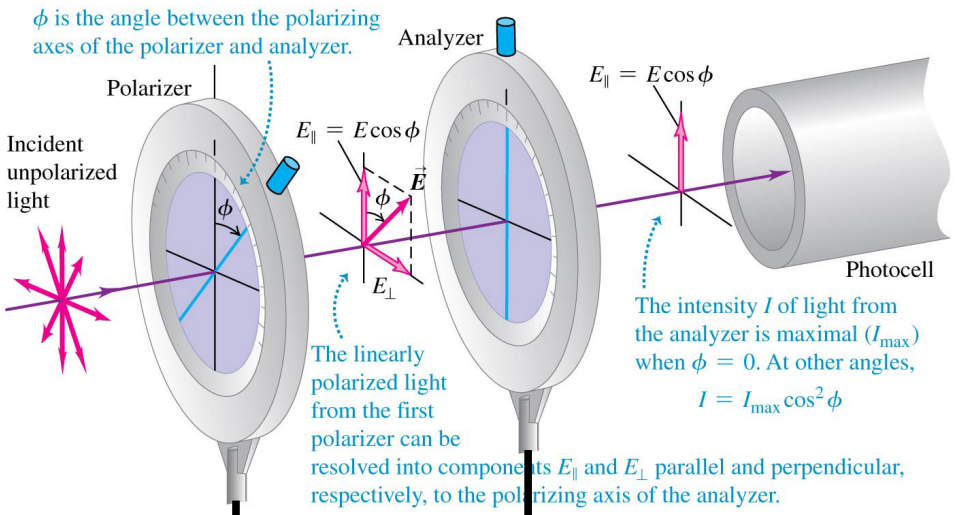
Maximum transmitted intensity

Angle between polarization axis of light and polarizing axis of analyzer

© 2016

Two Polarizers in Combination

$$\phi = 30°$$



I_0 $\quad$ $\frac{1}{2}I_0$ $\quad$ $I = \left(\frac{1}{2}I_0\right)\cos^2\phi = \frac{3}{8}I_0$

© 2016 Pearson Education, Ltd.

Question 24: Sunglass with polarized glass

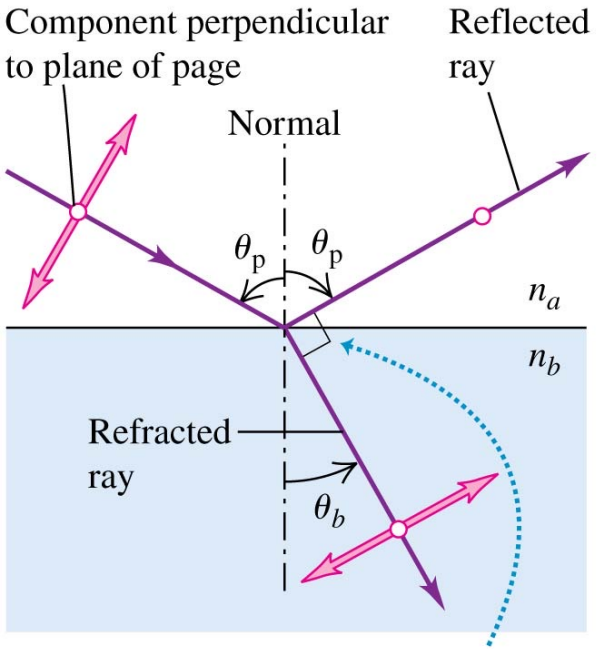
- To make a good sunglass with polarized glass, which types of glass should be used, vertically polarized or horizontally polarized?



© 2016 Pearson Education, Ltd.

Polarization by reflection

- Unpolarized light can be polarized, either partially or totally, by reflection.
- At one particular angle of incidence, called the **polarizing angle**, the light for which $\vec{E}$ lies in the plane of incidence is not reflected at all but is completely refracted.



When light strikes a surface at the polarizing angle, the reflected and refracted rays are perpendicular to each other and

$$\tan\theta_p = \frac{n_b}{n_a}$$

$$n_a \sin\theta_p = n_b \sin\theta_b = n_b \sin(90° - \theta_p) = n_b \cos\theta_p$$

Brewster's law for the polarizing angle:

Polarizing angle (angle of incidence for which reflected light is 100% polarized)

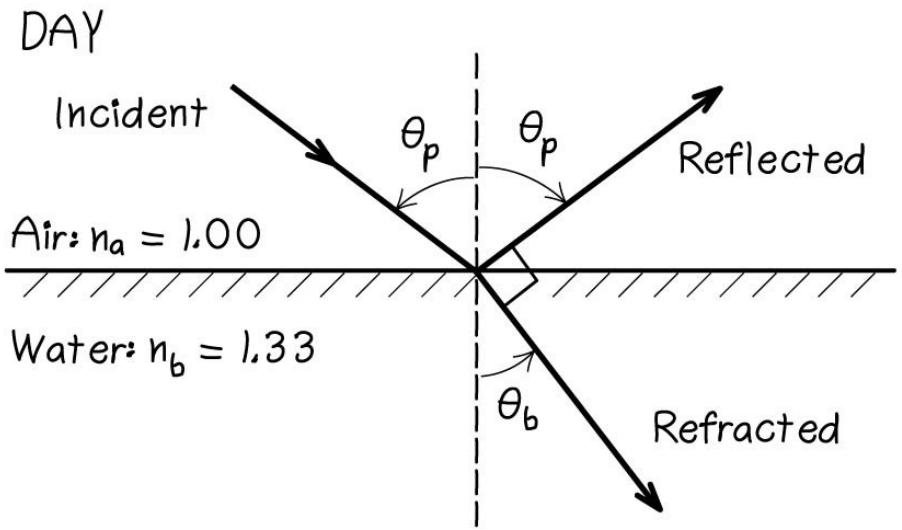
$$\tan\theta_p = \frac{n_b}{n_a}$$

n_b: Index of refraction of second material

n_a: Index of refraction of first material

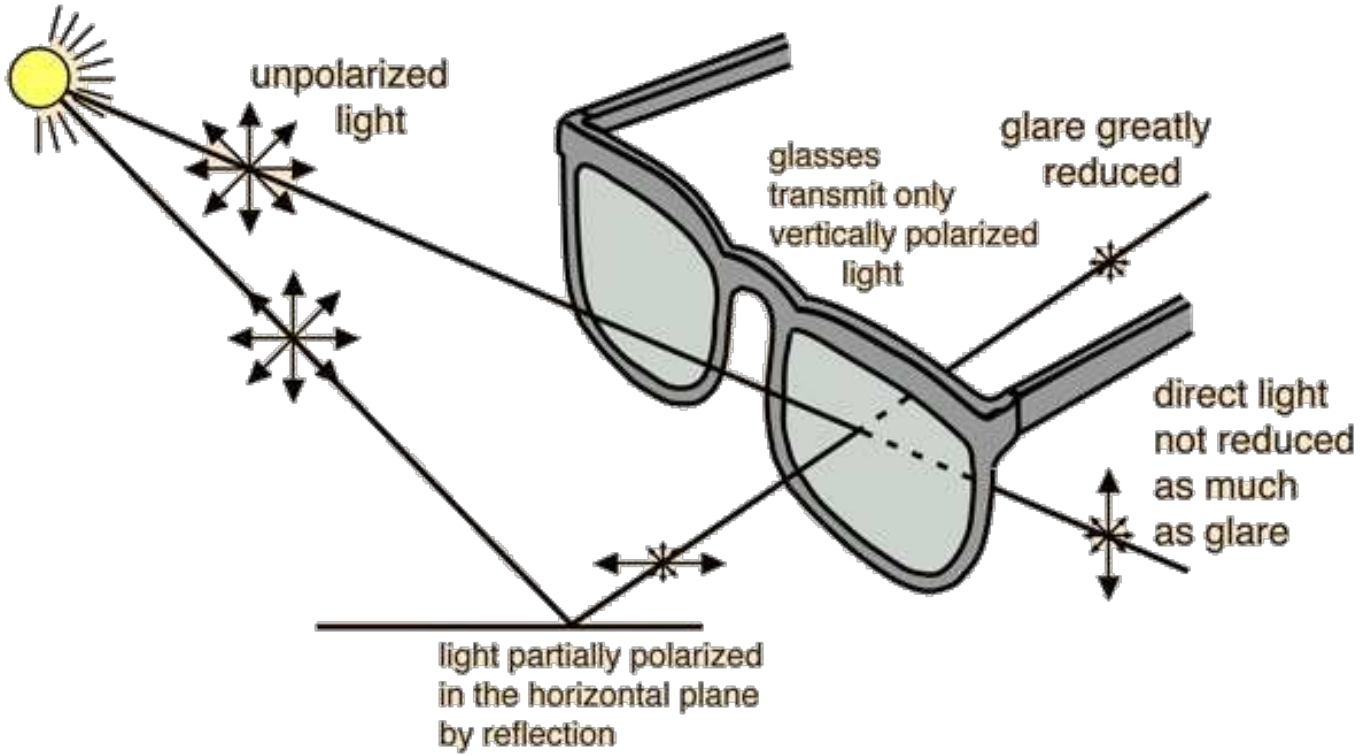
© 2016 Pearson Education, Ltd.

Reflection from a swimming pool's surface



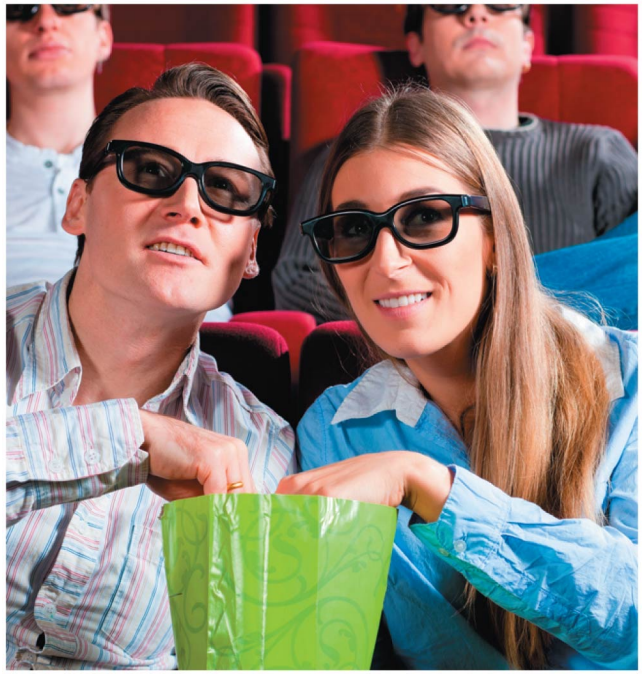
$$\theta_p = \arctan\frac{n_b}{n_a} = \arctan\frac{1.33}{1.00} = 53.1°$$

$$\theta_b = 90° - \theta_p = 90° - 53.1° = 36.9°$$



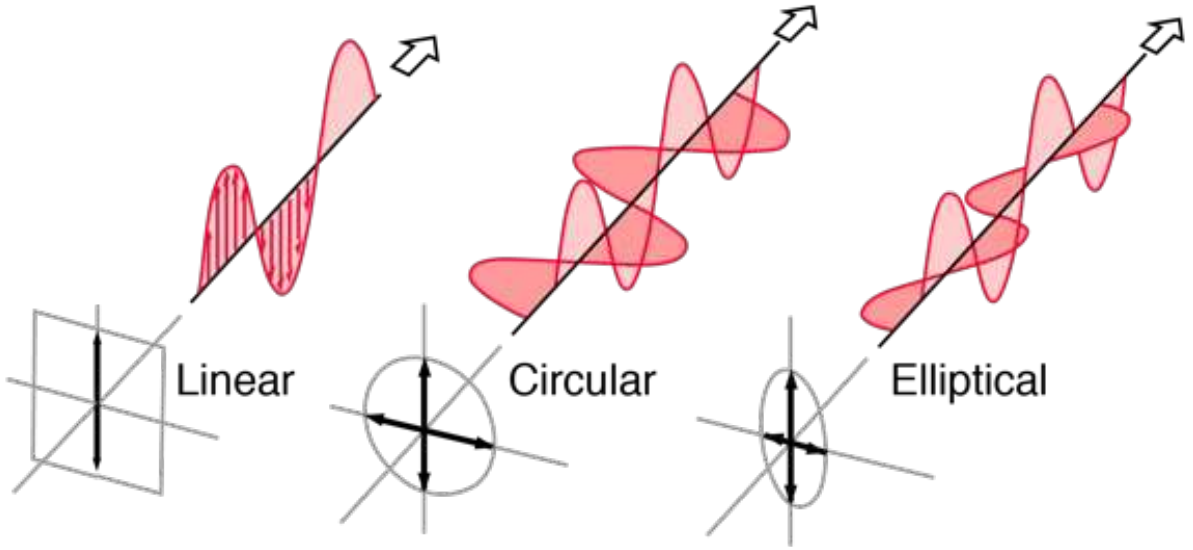
Circular polarization

- **Circular polarization** occurs when the $\vec{E}$ vector has a constant magnitude but rotates around the direction of propagation.
- When the wave is propagating toward you and the $\vec{E}$ vector appears to be rotating clockwise, it is called a **right circularly polarized** electromagnetic wave.
- If instead the $\vec{E}$ vector of a wave coming toward you appears to be rotating counterclockwise, it is called a **left circularly polarized** electromagnetic wave.
- The lenses of the special glasses you wear to see a 3-D movie are circular polarizing filters.



© 2016 Pearson Education, Ltd.

Circular polarization



Circularly polarized light

3D Images

© 2016 Pearson Education, Ltd.

Scattering of light

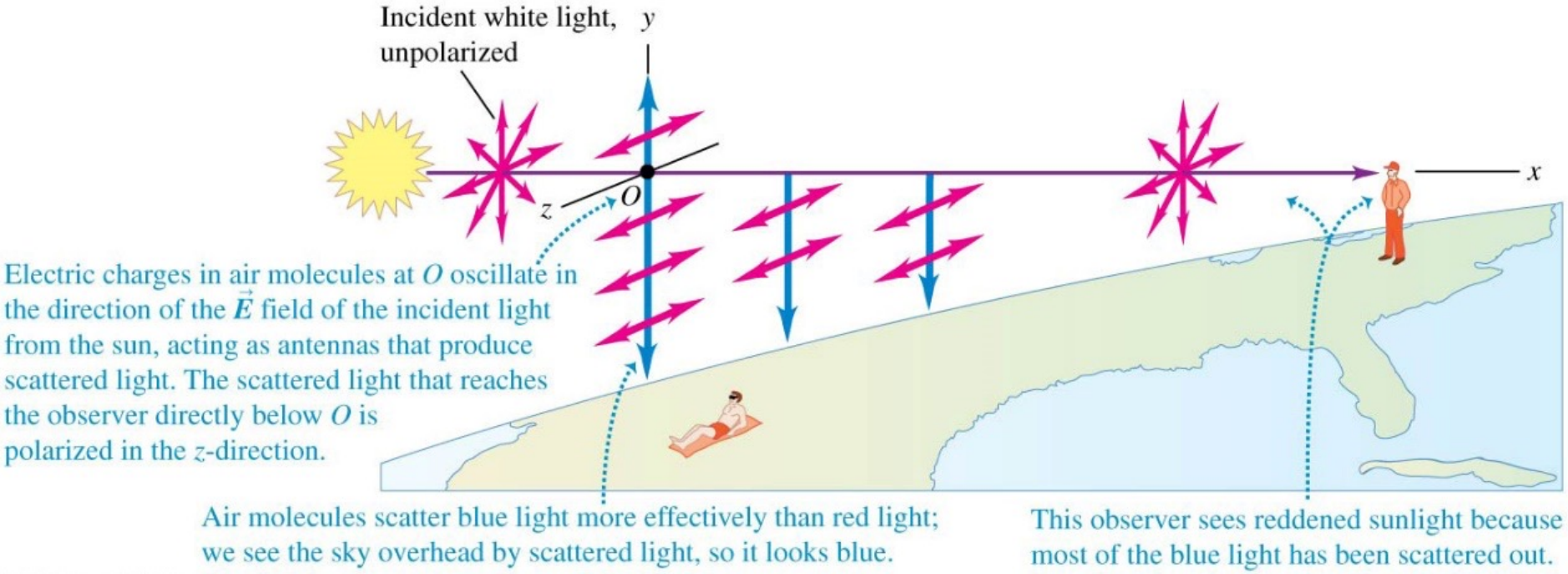
- When you look at the daytime sky, the light that you see is sunlight that has been absorbed and then re-radiated in a variety of directions.
- This process is called **scattering**.
- Light scattered by air molecules contains 15 times as much blue light as red, and that's why the sky is blue.





© 2016 Pearson Education, Ltd.

Scattering of light



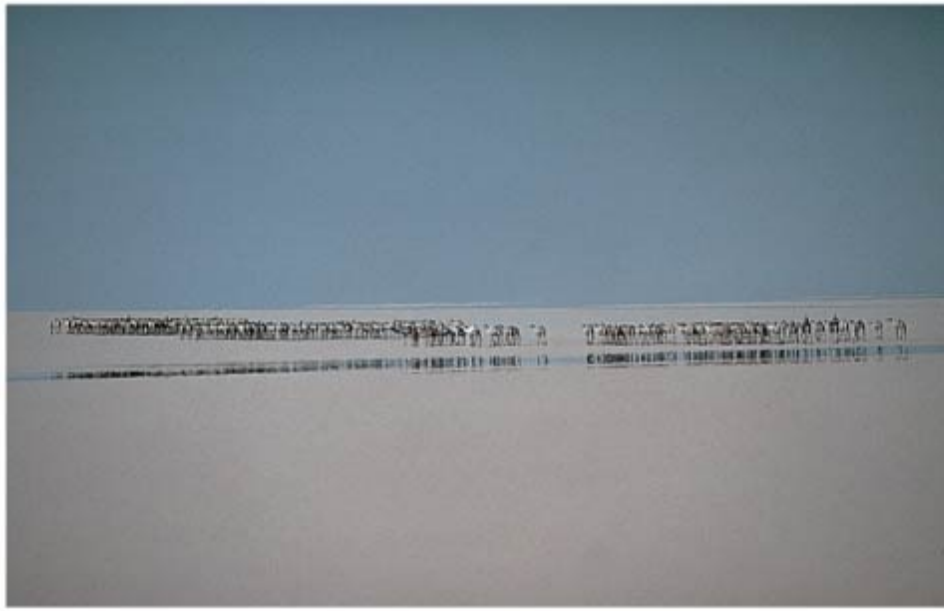
- Clouds contain a high concentration of suspended water droplets or ice crystals, which scatter light of all wavelengths equally, so the cloud looks white.

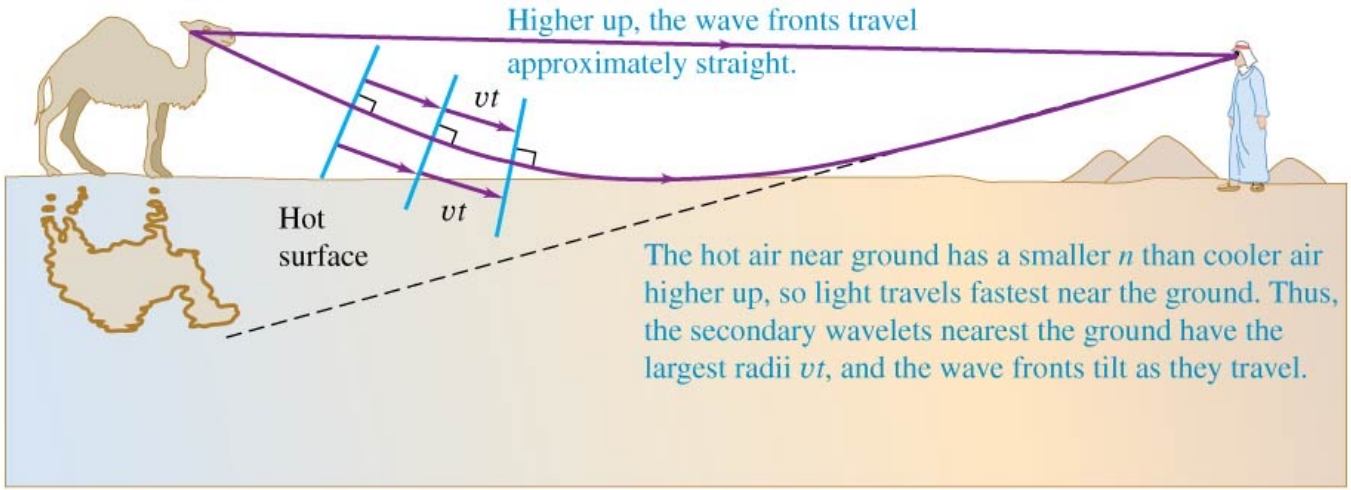


© 2016 Pearson Education, Ltd.

A mirage

- Mirages are an example of Huygens's principle.
- A thirsty traveler can interpret the apparent reflecting surface as a sheet of water.







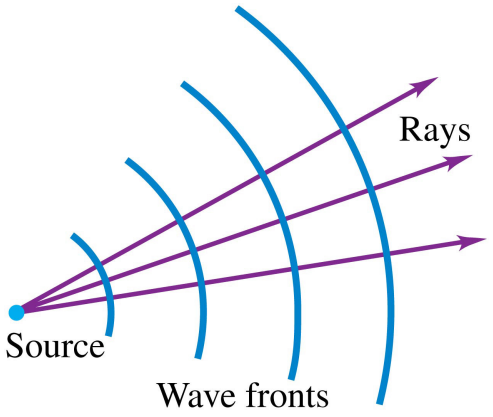
© 2016 Pearson Education, Ltd.

Summary

- Light and its properties

$$n = \frac{c}{v}$$

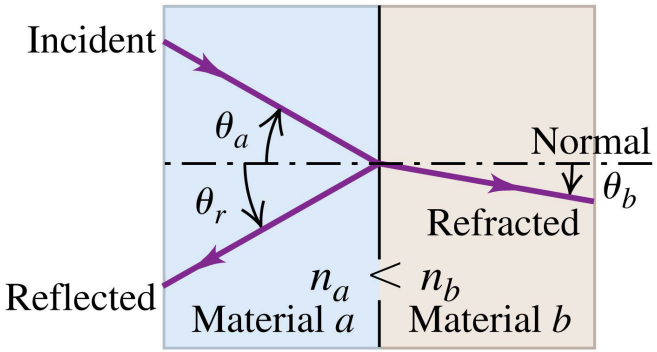
$$\lambda = \frac{\lambda_0}{n}$$



- Reflection and refraction

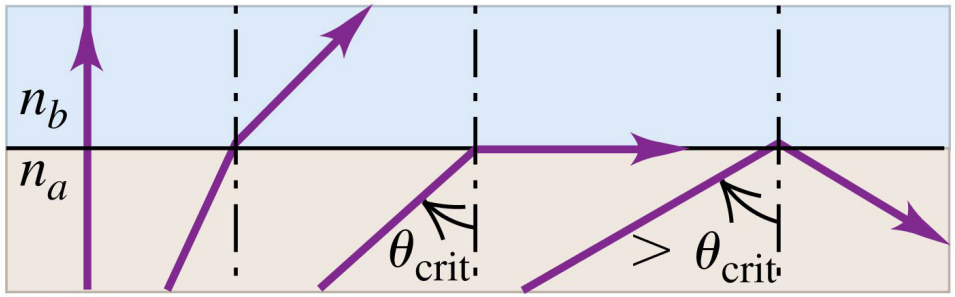
$\theta_r = \theta_a$
(law of reflection)

$n_a \sin\theta_a = n_b \sin\theta_b$
(law of refraction)



- Total internal reflection

$$\sin\theta_{\text{crit}} = \frac{n_b}{n_a}$$

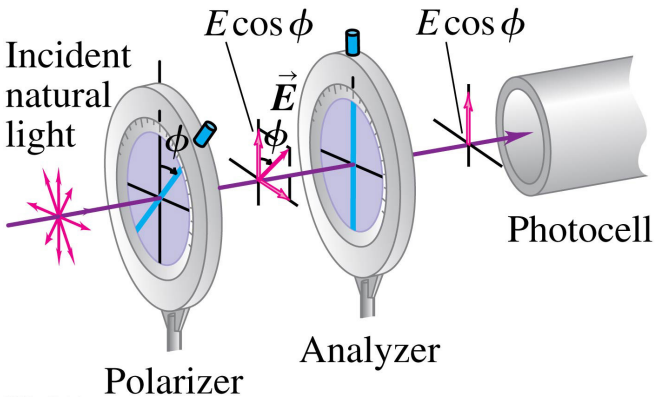


© 2016 Pearson Education, Ltd.

Summary

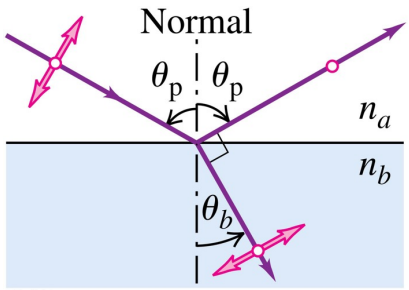
- Polarization of light

$$I = I_{\max} \cos^2 \phi$$

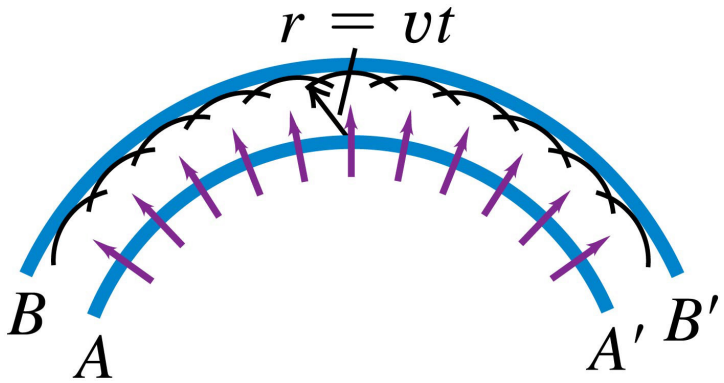


- Polarization by reflection

$$\tan\theta_p = \frac{n_b}{n_a}$$



- Huygens's principle



© 2016 Pearson Education, Ltd.